

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 271

THE QUINN OXY-DEFIB UNIT

55.5 mol to 888 g of oxygen

direct-connected to the engines — no bus in the path



100 people by 3 days — the 3 bridges are the evacuation points

CARBON ELECTRODES · THE REFILL DUTY · 250W BREAK-EVEN

THE WATER RESERVE IS THE AIR RESERVE

Emergency life support — v1.0 in force

GP-IM-271

Revision v1.0 — 24 August 2026

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SEATED LAW — EXTERNAL VET PENDING

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 271

PHASE 271: THE QUINN OXY-DEFIB UNIT — WALL-MOUNTED EMERGENCY ELECTROLYSIS OXYGEN — STATUS: CURRENT — SEATED LAW / EXTERNAL VET PENDING (CHARGE-TOPOLOGY AUDIT CORRECTION SEATED AT BIRTH, SITTING 151). A litre of water is a person-day: 55.5 mol H₂O makes 888 g of O₂ — a wall-mounted emergency electrolysis unit that turns the ship's water reserve into breathable days. The energy ledger is honest: about 6 kWh per litre, 250 W break-even — the unit runs direct-connected to the engines, because in an emergency you do not negotiate with a bus. The charge-topology audit correction was seated at birth (sitting 151): the wiring is direct, the physics priced. Electrodes: 'carbon or graphite is usually the choice for replaceable ones we can ma...' — replaceable, makeable, in the file's own shop. The reserve doctrine: '100 people by 3 days as our emergency water' — 300 person-days on the wall; 'the 3 bridges will be the emergency evacuation points' — the units mount where the crowd gathers. The refill duty (sitting 155): a unit that has breathed for someone gets refilled that watch — the defib cabinet logic, for air.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-271, v1.0 — 24 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the unit law as seated (contents line, the unit in crayon — the litre ledger, the energy price, the direct connection, the electrodes, the reserve, the bridges — the corrections and the refill duty, entire) — §2 the physics, graded — §3 the system on the wall — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 271: **The Oxy-Defib Unit — Wall-Mounted Emergency Electrolysis Oxygen**. A litre of water is a person-day at rest (888 g O₂; ~6 kWh/L at the 1.8–2.5 V low-

voltage high-current cell band; 0.23–0.29 kW sustains one person indefinitely). Direct-connected to the engines; 1 IBC covers 100 people for 3 days with 3.3× margin; commercial metal electrodes ship-spec with onboard carbon/graphite fabrication; bottle-refill compressor duty for repair crews.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: emergency breathing oxygen anywhere on the ship, from any water access point, independent of batteries and of lines a hull breach can cut — wall-mounted electrolysis units direct-connected to the orbital engines and cylinder engine banks, because whatever the cause is, nothing stopped objects in motion.'

THE INVENTION (THE AUTHOR, VERBATIM, 10 August 2026) (VERBATIM)

"if you are running out of oxygen, the amount in water is massive, you simply need an anode a cathode a hydrogen vacuum line out and a high voltage charge, transformers can provide the charge level, a litre bottle of water with the unit will do the rest, because whatever the cause is, nothing stopped objects in motion, and our ships electrical power is objects in motion, the orbital engines and cylinder engine banks."

- **THE CHARGE TOPOLOGY — AUDIT CORRECTION (seated at the invention's birth, sitting 151):** the chemistry floor is 1.23 V; practical electrolysis cells run 1.8–2.5 V. This is a LOW-VOLTAGE, HIGH-CURRENT machine — welder class, step-down transformer — NOT high voltage. The author's 'high voltage charge' wording is preserved verbatim above per doctrine; the buildable topology is high CURRENT at cell voltage, and the transformer steps DOWN.

THE ORDERED MATH (SITTING 151) (VERBATIM)

The author's instruction, verbatim: "i will not quote random numbers i will let kimi give you voltage over time for output from a lite of water". The math, as ordered and seated:

- **ONE LITRE = ONE PERSON-DAY AT REST.** 1 litre of water = 55.5 mol → 27.75 mol O₂ = 888 g O₂ ≈ one person's resting day (~0.84 kg/day consumption), plus 112 g H₂ out the vacuum line. Charge passed: 10.71 megacoulombs (4 faradays per O₂).
- **THE ENERGY BILL.** At the practical cell band the litre costs ~6 kWh. 250 W is break-even for one resting person; 0.23–0.29 kW sustains one person indefinitely; a 5 kW unit makes a person-day in about 72 minutes.

THE BUILD SPEC (THE AUTHOR, 10 August 2026, SITTING 153) (VERBATIM)

- **ELECTRODES, BOTH KINDS, VERBATIM:** "carbon or graphite is usually the choice for replaceable ones we can make for the anode and cathode, but for the ship we will use the commercial metal ones and have specs and testing making the other on board." Ship units: commercial metal electrodes, specified and tested. Onboard fabrication capability: replaceable carbon/graphite electrodes, made

and tested on board.

- **THE EMERGENCY WATER BUDGET, VERBATIM:** "so we need to ensure 100 people by 3 days as our emergency water so 1 IBC is enough." At the ordered math: $100 \text{ people} \times 3 \text{ days} = 300 \text{ person-days} = 300 \text{ litres}$; one 1,000-litre IBC $\approx 3.3\times$ margin.
- **THE DEPLOYMENT LAW, VERBATIM:** "the 3 bridges will be the emergency evacuation points as they are evenly spread across the ship and large enough for 80 people. thus we have just required a 4 toilet ensuite to every bridge." The evacuation routes between bridges are the SMOTs — mayernicks run both ways on a polarity switch on the sled; the demo lifted 6 times its own weight up and over a vertical loop (sitting 154).
- **THE INSTALLATION, VERBATIM (sitting 151):** "Wall mounted, direct connect to the engines in case it has been a battery issue or cut lines from a series of hull breaches."
- **THE REFILL DUTY (sitting 155):** every unit connects to a bottle refill device/compressor so repair crews working hull breaches can refill mask bottles — the air doctrine's mask-bottles-in-all-rooms law (sitting 150). Hardware is available-spec: commercial dive-shop breathing-air compressor units, oil-free, filtered, moisture-separated (sittings 155–156).

ORIGIN OF THOUGHT — PROVENANCE (VERBATIM)

Invented by the author, 10 August 2026, out of the nitrogen-myth kill: we breathe out all the nitrogen we breathe in — nitrogen is a dilution necessity, but because we breathe it straight back out, only the oxygen fraction ever needs replacing in an emergency, and the oxygen in water is massive. The author's whiskey-glass demonstration and the full doctrine are seated verbatim in sitting 151. Phase number 271 assigned 11 August 2026 on the author's order: "seat the defib" (next unused number).

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE LITRE LEDGER IS THE HEART: 55.5 mol of water per litre, and every mole of H_2O gives half a mole of O_2 — 888 g of oxygen, a person-day, from a litre. The reserve is not tanks of gas; it is the ship's water, one phase-change away from breath.

THE ENERGY PRICE IS PAID IN THE OPEN: about 6 kWh per litre — electrolysis is not cheap, and the phase says so; 250 W break-even sets the trickle rate. Emergency oxygen is bought with energy, and the ledger is priced before the emergency.

DIRECT-CONNECTED TO THE ENGINES: in an emergency you do not negotiate with a bus — the unit draws from the engines' own supply, the charge-topology correction seated at birth (sitting 151) so the wiring matches the physics.

THE ELECTRODES ARE SHOP-MAKEABLE: 'carbon or graphite is usually the choice for replaceable ones we can ma...' — replaceable, makeable in the file's own shop, no exotic catalyst supply chain between the crew and their air.

THE RESERVE IS SIZED IN PEOPLE: '100 people by 3 days as our emergency water' — 300 person-days on the wall; and 'the 3 bridges will be the emergency evacuation points' — the units mount where the crowd gathers, because air at the muster point is the doctrine.

THE REFILL DUTY IS THE DEFIB LOGIC (sitting 155): a unit that has breathed for someone gets refilled that watch — like the defib cabinet, the box is only rescue if it is full. The duty is named, assigned, and drilled.

§3 — THE SYSTEM ON THE WALL

- **The ledger:** a litre is a person-day — 55.5 mol, 888 g O₂.
- **The price:** about 6 kWh per litre; 250 W break-even — paid in the open.
- **The feed:** direct-connected to the engines — no bus negotiation in an emergency (sitting 151).
- **The electrodes:** carbon or graphite, replaceable, shop-makeable.
- **The reserve:** 100 people by 3 days — 300 person-days of emergency water.
- **The posts:** the 3 bridges are the evacuation points — the units mount where the crowd gathers; the refill duty runs that watch (sitting 155).

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- Sitting 151 seated the charge-topology correction at birth — the audit caught the wiring before the wall did; the correction is part of the phase's record, in the open.
- Sitting 155 seated the refill duty — the defib-cabinet logic applied to air: used means refilled, that watch, named and drilled.
- The water doctrine is the family: 259's shield tanks and 258's foundry water are the same ledger — water is cargo, shield, and breath, priced one way across the file.
- The engines' direct feed ties to the bank doctrine: 264/269's power — the emergency draw is priced against the same supply the station runs on.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- 1. Build the cell: carbon or graphite electrodes, replaceable — the shop-makeable stack.
- 2. Wire it direct: the engines' own supply, the sitting-151 topology — no bus in the path.
- 3. Mount at the muster points: the 3 bridges first — air where the crowd gathers.
- 4. Fill the reserve: 100 people by 3 days — 300 person-days of emergency water, logged.
- 5. Drill the duty: used means refilled that watch (sitting 155) — the drill card on the cabinet door.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phases 264/269 (the bank):** the feed — the emergency draw priced against the engines' supply.
- **Phase 259 (the shield tanks):** the water ledger — cargo, shield, and breath from the same stock.
- **Phase 258 (water from rocks):** the source — the foundry refills the reserve on the Moon.
- **Phase 246 (the medical band):** the cousin — emergency medicine on the wall, one for the heart, one for the breath.

§8 — DO-NOT-BUILD

- Do NOT route it through the bus — direct-connected to the engines; emergency air does not queue.
- Do NOT spec exotic electrodes — carbon or graphite, replaceable, shop-makeable; the supply chain is the shop.
- Do NOT undersize the reserve — 100 by 3 days is the floor; the ledger is counted in people.
- Do NOT mount it away from the muster — the 3 bridges are the evacuation points; air goes where the crowd goes.
- Do NOT leave a used unit empty — refilled that watch; a defib cabinet logic, for air.

§9 — OWED

OWED: the bench list, whole — cell stack spec and electrode life; the 250 W trickle profile versus the 6 kWh sprint; mounting hardware at the three bridges; the reserve gauge and log; the drill card. And the external vet, when it lands.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-271, v1.0 — CURRENT — SEATED LAW / EXTERNAL VET PENDING (CHARGE-TOPOLOGY AUDIT CORRECTION SEATED AT BIRTH, SITTING 151), seated law, master v2.1 (numerical print order). Built 24 August 2026, twelfth of the 20-chapter 'you get the next 20 ready' shift (the author's order, 24 August 2026, seated in sitting 623 — 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618, 622 and 623): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20' / 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 271